Specification Packaging Material

E-62843-01, LEAFLET GLUCIENT XR EXTENDED RELEASE TABLET MALAYSIA

Author(s):

06.03.2025, Gina Sonya, Artwork Management Supervisor

Reviewed by:

07.03.2025, Riski Amanda Tambunan, Artwork Management Manager

Approved by:

Effective date:

170 mm

1.5 mm 9 mm

15 mm

GLUCIENT XR

750 mg and 1000 mg Extended-release Tablet

Dexa Medica

Pharmacode: 8431

Compositions:
GLUCIENT XR 750 MG EXTENDED-RELEASE TABLET
Each extended-release tablet contains
Metformin HCl 750 mg

GLUCIENT XR 1000 MG EXTENDED-RELEASE TABLET
Each extended-release tablet contains
Metformin HCl 1000 mg

Product Descriptions:
GLUCIENT XR 750 MG EXTENDED-RELEASE TABLET
White tablet, shallow convex, dimension 20 x 9 mm, with marking "SR1" on one side and without marking on the other side.

GLUCIENT XR 1000 MG EXTENDED-RELEASE TABLET
White tablet, shallow convex, dimension 22.25 x 9 mm, with marking "SR2" on one side and without marking on the other side.

Pharmacodynamics:
Metformin is a biguanide with antihyperglycemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycemia.
Metformin may act via 3 mechanisms:
- Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis.
- In muscle, by increasing insulin sensitivity, improving peripheral glucose uptake and utilization.
- And, delay of intestinal glucose absorption.

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase. Metformin increases the transport capacity of all types of membrane glucose transporters (GLUT).

Pharmacokinetics:
Absorption
GLUCIENT XR Extended-release Tablet bioequivalence study was conducted by single dose under fasting conditions. Following the single dose administered under fasting condition, C_{max} is achieved with a median value of 8 hours and a range of 1.50 hours to 10.07 hours, while AUC_{0-t}, and C_{max} are similar to the reference drug.
Mean metformin absorption from the extended-release formulation is not altered by meal composition.
No accumulation is observed after repeated administration of up to 2,000 mg metformin as extended-release tablets.

Distribution
Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean volume of distribution (V_d) ranged between 63–276 l.

Metabolism
Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Excretion
Renal clearance of metformin is >400 ml/minute, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

Characteristics in specific groups of patients
Renal impairment
The available data in subjects with moderate renal insufficiency are scarce and no reliable estimation of the systemic exposure to metformin in this subgroup as compared to subjects with normal renal function could be made. Therefore, the dose adaptation should be made upon clinical efficacy/tolerability considerations.

Indications:
- Reduction in the risk or delay of the onset of type 2 diabetes mellitus in adult, overweight patients with Impaired Glucose Tolerance and/or Impaired Fasting Glucose, and/or increased HbA1c who are:
  - at high risk for developing overt type 2 diabetes mellitus and
  - still progressing towards type 2 diabetes mellitus despite implementation of intensive lifestyle change for 3 to 6 months.

  Treatment with Glucient XR Extended-release Tablet must be based on a risk score incorporating appropriate measures of glycemic control and including evidence of high cardiovascular risk.
  Lifestyle modifications should be continued when metformin is initiated, unless the patient is unable to do so because of medical reasons.
- Treatment of type 2 diabetes mellitus in adults, when dietary management and exercise alone does not result in adequate glycemic control. Glucient XR Extended-release Tablet may be used as monotherapy or in combination with other oral antidiabetic agents, or with insulin.

For Glucient XR 750 mg Extended-release Tablet: A reduction of diabetic complications has been shown in overweight type 2 diabetic patients treated with immediate-release metformin as first-line therapy after diet failure.

Recommended Dose:
Adults with normal renal function (GFR ≥90 ml/minute)
Reduction in the risk or delay of the onset of type 2 diabetes:
- Metformin should only be considered where intensive lifestyle modifications for 3 to 6 months have not resulted in adequate glycemic control.
- The therapy should be initiated with one tablet metformin hydrochloride extended-release 500 mg once daily with the evening meal.
- After 10 to 15 days dose adjustment on the basis of blood glucose measurements is recommended (OGTT and/or FPG and/or HbA1c values to be within the normal range). A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose is 2,000 mg once daily with the evening meal.
- It is recommended to regularly monitor (every 3–6 months) the glycemic status (OGTT and/or FPG and/or HbA1c value) as well as the risk factors to evaluate whether treatment needs to be continued, modified, or discontinued.
- A decision to reevaluate therapy is also required if the patient subsequently implements improvements to diet and/or exercise, or if changes to the medical condition will allow increased lifestyle interventions to be possible.

Monotherapy and combination with other oral antidiabetic agents:
- For patients new to metformin hydrochloride, the usual starting dose of metformin hydrochloride extended-release tablet is 500 mg or 750 mg once daily given with the evening meal. After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose is 2,000 mg once daily with the evening meal.
- In patients already treated with metformin tablets, the starting dose of Glucient XR Extended-release Tablet should be equivalent to the daily dose of metformin immediate-release tablets. In patients treated with metformin hydrochloride at a dose above 2,000 mg daily, switching to Glucient XR Extended-release Tablet is not recommended.
- If glycemic control is not achieved on 2,000 mg metformin hydrochloride extended-release once daily, patients may be switched to standard metformin hydrochloride immediate-release to a maximum dose of 3,000 mg daily.
- If transfer from another oral antidiabetic agent is intended: discontinue the other agent and initiate Glucient XR Extended-release Tablet at the dose indicated above.

Combination with insulin:
Metformin hydrochloride and insulin may be used in combination therapy to achieve better blood glucose control. The usual starting dose of metformin hydrochloride extended-release is one tablet 500 mg or 750 mg once daily with the evening meal, while insulin dosage is adjusted on the basis of blood glucose measurements. After titration, switch to Glucient XR 1000 mg Extended-release Tablet should be considered.

Elderly:
Due to the potential for decreased renal function in elderly patients, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary.
Benefit in the reduction of risk or delay of the onset of type 2 diabetes mellitus has not been established in patients 75 years and older and metformin initiation is therefore not recommended in these patients.

Renal impairment:
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3–6 months.

<table>
<tr><th>GFR (ml/minute)</th><th>Total maximum daily dose</th><th>Additional considerations</th></tr>
<tr><td>60–89</td><td>2,000 mg</td><td>Dose reduction may be considered in relation to declining renal function.</td></tr>
<tr><td>45–59</td><td>2,000 mg</td><td rowspan="2">Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin.
The starting dose is at most half of the maximum dose.</td></tr>
<tr><td>30–44</td><td>1,000 mg</td></tr>
<tr><td><30</td><td>-</td><td>Metformin is contraindicated.</td></tr>
</table>

Pediatric population:
In the absence of available data, Glucient XR Extended-release Tablet should not be used in children.

Route of Administration:
For oral use.

Contraindications:
- Hypersensitivity to metformin or to any of the excipients.
- Diabetic precoma.
- Severely reduced renal function (GFR <30 ml/minute).
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis).
- Acute conditions with the potential to alter renal function such as:
  - dehydration,
  - severe infection,
  - shock.
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as:
  - decompensated heart failure,
  - respiratory failure,
  - recent myocardial infarction,
  - shock.
- Hepatic insufficiency, acute alcohol intoxication, alcoholism.

62843-01XX-YY
Last revised: dd/mm/yyyy

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Color : Black	**Font type (Product name)** : Gothic 725 Bold BT **(Redactional)** : Arial **Font size (Product name)** : 20 pt **(Redactional)** : 6 pt ***Double side leaflet** ***Page 1***

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Pharmacode: 8431

Warnings and Precautions:

Lactic acidosis:
Lactic acidosis is a very rare, but serious, metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.
In case of dehydration (severe diarrhea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.
Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics, and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting, and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.
Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterized by acidotic dyspnea, abdominal pain, muscle cramps, asthenia, and hypothermia followed by coma. In case of suspected symptoms, the patients should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/l), and an increased anion gap and lactate/pyruvate ratio.

Renal function:
GFR should be assessed before treatment initiation and regularly thereafter (see section **Recommended Dose**). Metformin is contraindicated in patients with GFR <30 ml/minute and should be temporarily discontinued in the presence of conditions that alter renal function (see section **Contraindications**).

Cardiac function:
Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, metformin may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, metformin is contraindicated.

Elderly:
Due to the limited therapeutic efficacy data in the reduction of risk or delay of type 2 diabetes in patients 75 years and older, metformin initiation is not recommended in these patients.

Administration of iodinated contrast agents:
Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been reevaluated and found to be stable.

Surgery:
Metformin must be discontinued at the time of surgery under general, spinal, or epidural anesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been reevaluated and found to be stable.

Other precautions:
- All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.
- Metformin alone never causes hypoglycemia, although caution is advised when it is used in combination with insulin or other oral antidiabetics (e.g. sulfonylureas or meglitinides).

Effects on ability to drive and use machines
Metformin monotherapy does not cause hypoglycemia and therefore has no effect on the ability to drive or to use machines. However, patients should be alerted to the risk of hypoglycemia when metformin is used in combination with other antidiabetic agents (e.g. sulfonylureas, insulin, or meglitinides).

Interactions with Other Medicaments:
Concomitant use not recommended
- Alcohol
  Acute alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in case of fasting, malnutrition, or hepatic impairment.
- Iodinated contrast media
  Metformin must be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been reevaluated and found to be stable.

Combinations requiring precautions for use
Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclooxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists, and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.
- Medicinal products with intrinsic hyperglycemic activity (e.g. glucocorticoids (systemic and local routes) and sympathomimetics).
  More frequent blood glucose monitoring may be required, especially at the beginning of treatment. If necessary, adjust the metformin dosage during therapy with the other drug and upon its discontinuation.
- Organic cation transporters (OCT)
  Metformin is a substrate of both transporters OCT1 and OCT2.
  Coadministration of metformin with:
  - Inhibitors of OCT1 (such as verapamil) may reduce efficacy of metformin.
  - Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.
  - Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprim, vandetanib, isavuconazole) may decrease the renal elimination of metformin and this lead to an increase in metformin plasma concentration.
  - Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.
  Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.

Pregnancy and Lactation:
Pregnancy:
Uncontrolled diabetes during pregnancy (gestational or permanent) is associated with increased risk of congenital abnormalities and perinatal mortality.
When the patient plans to become pregnant and during pregnancy, it is recommended that impaired glycemic control or diabetes are not treated with metformin. For diabetes, it is recommended that insulin should be used to maintain blood glucose levels as close to normal as possible to reduce the risk of malformations of the fetus.

Lactation:
Metformin is excreted into human breast milk. No adverse effects were observed in breastfed newborns/infants. However, as only limited data are available, breastfeeding is not recommended during metformin treatment. A decision on whether to discontinue breastfeeding should be made, taking into account the benefit of breastfeeding and the potential risk to adverse effect on the child.

Fertility:
Fertility of male or female rats was unaffected by metformin when administered at doses as high as 600 mg/kg/day, which is approximately three times the maximum recommended human daily dose based on body surface area comparisons.

Adverse Effects:
Metabolism and nutrition disorders:
Very rare:
- Lactic acidosis.
- Decrease of vitamin B_{12} absorption with decrease of serum levels during long-term use of metformin. Consideration of such an etiology is recommended if a patient presents with megaloblastic anemia.

Nervous system disorders:
Common: taste disturbance.

Gastrointestinal disorders:
Very common: gastrointestinal disorders such as nausea, vomiting, diarrhea, abdominal pain, and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. A slow increase of the dose may also improve gastrointestinal tolerability.

Hepatobiliary disorders:
Very rare: isolated reports of liver function tests abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders:
Very rare: skin reactions such as erythema, pruritus, urticaria.

Overdose and Treatment:
Hypoglycemia has not been seen with metformin doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose or concomitant risks of metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and metformin is hemodialysis.

Presentations and Registration Numbers:
GLUCIENT XR 750 MG EXTENDED-RELEASE TABLET: PVC/PVDC coated RF - Alu Foil. 2 x 14 tablets in a carton; MAL24046024ACZ
GLUCIENT XR 1000 MG EXTENDED-RELEASE TABLET: PVC/PVDC coated RF - Alu Foil. 4 x 15 tablets in a carton; MAL24046025ACZ

ON MEDICAL PRESCRIPTION ONLY.

STORE AT TEMPERATURES BELOW 30ºC. PROTECT FROM LIGHT AND MOISTURE.

Manufactured by
PT Ferron Par Pharmaceuticals
Kawasan Industri Jababeka I
Jl. Jababeka VI Blok J3, Cikarang
Bekasi-Indonesia

For
PT Dexa Medica
Jl. Jenderal Bambang Utoyo No. 138
Palembang 30115-Indonesia

Product Registration Holder
Pharmaforte (Malaysia) Sdn. Bhd.
2, Jalan PJU 3/49, Sunway Damansara,
47810 Petaling Jaya,
Selangor Darul Ehsan, Malaysia

Date of Revision: 04 March 2025

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Fiber direction

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Black	**(Redactional) :** Arial	
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	(Redactional) : 6 pt	
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No.	Parameters	Specification
1.	Material	HVS (Lihat CoA / See CoA)
2.	Gramatur / Grammature (gsm)	54 – 66 gsm
3.	Dimensi panjang / Length dimension (mm)	299 – 301 mm
4.	Dimensi lebar / Width dimension (mm)	169 – 171 mm
5.	Arah serat / Fiber direction	Vertikal / Vertical
6.	Warna / Colour	Lihat proof print / See proof print
7.	Redaksional / Redactional	Lihat proof print / See proof print

HISTORY

Revision No.	Revision Date	Details	Reason
00	-	Dokumen baru	ProdCom000946/1
01	28-Jun-18	Perubahan storage condition menghilangkan “Protect from Light”	SFP-A-10054
02	11-Dec-18	Penambahan strength 750 mg	ProdCom001075/1
03	25-Jun-19	- Penambahan 500 pada nama brand - Penghilangan strength 750 mg - Perubahan redaksional	CMCOthRec050577/1
04	30-Sep-19	Perubahan redaksional	CMCOthRec053936/1
05	11-Dec-19	Perubahan redaksional	CMCOthRec056755/1 CMCOthRec056756/1
06	09 Aug 2021	Perubahan redaksional sesuai PI dari RA	- CMCOthRec076004/1 - CMCOthRec075998/1
07	27 Dec 2021	- Perubahan product registration holder menjadi “Rakss Solutions” - Perubahan redaksional sesuai PI terbaru	- CMCOthRec081056/1 - CMCOthRec081054/2 - CMCOthRec075998/2
08	15 Mei 2023	Perubahan redaksional sesuai PI terbaru	- CMCOthRec102530/1 - CMCOthRec075998/3
09	02 Aug 2023	Tambahan data terkait perubahan redaksional sesuai PI terbaru	- CMCOthRec107152/1 - CMCOthRec075998/4
10	15 Jan 2024	Tambahan data terkait perubahan redaksinaol sesuai PI terupdate	- CMCOthRec075998/5 - CMCOthRec115440/1
11	21 Jan 2025	Perubahan dokumen menjadi general use, dengan detail : - Penambahan tanggal revisi terakhir - Penambahan approval site - Penambahan nomor registrasi Malaysia	- CMCOthRec134071/1 - CMCOthRec134073/1
12	06 Mar 2025	- Revisi item number menjadi E-62843-01 - Perubahan “Product Registration Holder” - Perubahan “Date of Revision”	- CMCOthRec136596/1 - CMCOthRec075998/6